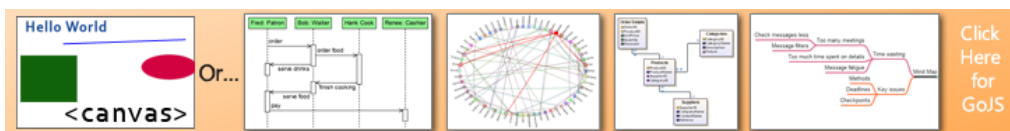


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Detecting cycles in an adjacency matrix



Let A be the adjacency matrix for the graph $G = (V, E)$. $A(i, j) = 1$ if the nodes i and j are connected with an edge, $A(i, j) = 0$ otherwise.

My objective is the one of understanding whether G is acyclic or not. A cycle is defined in the following way:

- i and j are connected: $A(i, j) = 1$
- j and k are connected: $A(j, k) = 1$
- k and i are connected: $A(k, i) = 1$

I have implemented a solution which navigates the matrix as follows:

- Start from an edge (i, j)
- Select the set o of edges which are outgoing from j , i.e., all the 1s in the j -th row of A
- Navigate o in a DFS fashion
- If one of the paths generated from this navigation leads to the node i , then a cycle is detected

Obviously this solution is very slow, since I have to evaluate all the paths in the matrix. If A is very big, the required overhead is very huge. I was wondering whether there is a way of navigating the adjacency matrix so as to find cycles without using an expensive algorithm such as DFS.

I would like to implement my solution in MATLAB.

Thanks in advance,

Eleanore.

[matlab](#) [graph](#) [matrix](#) [graph-algorithm](#)

asked May 8 '13 at 8:40

[Eleanore](#)

545 5 16

7 Answers

Based on the observation of [Danil](#), you need to compute A^n , a slightly more efficient way of doing so is

```
n = size(A,1);
An = A;
for ii = 2:n
    An = An * A; % do not re-compute A^n from scratch
    if trace(An) ~= 0
        fprintf(1, 'got cycles\n');
    end
end
```

answered May 8 '13 at 10:32

[Shai](#)

31k 12 38 76

Quite simple but not too efficient. n matrix multiplications are quite a few. – [Dukeling](#) May 8 '13 at 11:21

1 @Dukeling better than n matrix powers, though... – [Shai](#) May 8 '13 at 11:27

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If A is the adjacency matrix of the directed or undirected graph G , then the matrix A^n (i.e., the matrix product of n copies of A) has following property: the entry in row i and column j gives the number of (directed or undirected) walks of length n from vertex i to vertex j .

E.g. if for some integer n matrix A^n contain at least one non-zero diagonal entry, than graph has cycle of size n .

Most easy way check for non-zero diagonal elements of matrix is calculate matrix $\text{trace}(A) = \text{sum}(\text{diag}(A))$ (in our case elements of matrix power will be always non-negative).

Matlab solution can be following:

```
for n=2:size(A,1)
    if trace(A^n) ~= 0
        fprintf('Graph contain cycle of size %d', n)
        break;
    end
end
```

edited May 8 '13 at 10:34

answered May 8 '13 at 9:40



Danil Asotsky

1,183 4 9 23

You don't have to re-compute A^n at each iteration. see stackoverflow.com/a/16438379/1714410. – Shai May 8 '13 at 10:36

A reference about that claim of yours. – Dukeling May 8 '13 at 11:38

I came across this question when answering this [math.stackexchange](http://math.stackexchange.com) question. For future readers, I feel like I need to point out (as others have already) that Danil Asotsky's answer is incorrect, and provide an alternative approach. The theorem Danil is referring to is that the (i,j) entry of A^k counts the number of *walks* of length k from i to j in G . The key thing here is that a *walk* is allowed to repeat vertices. So even if a diagonal entries of A^k is positive, each walk the entry is counting may contain repeated vertices, and so wouldn't count as a cycle.

Counterexample: A path of length 4 would contain a 4-cycle according to Danil's answer (not to mention that the answer would imply $P=NP$ because it would solve the Hamilton cycle problem).

Anyways, here is another approach. A graph is acyclic if and only if it is a forest, i.e., it has c components and exactly $n-c$ edges, where n is the number of vertices. Fortunately, there is a way to calculate the number of components using the [Laplacian matrix](#) L , which is obtained by replacing the (i,i) entry of $-A$ with the sum of entries in row i of A (i.e., the degree of vertex labeled i). Then it is known that the number of components of G is $n - \text{rank}(L)$ (i.e., the multiplicity of 0 as an eigenvalue of L).

So G has a cycle if and only if the number of edges is at least $n - (n - \text{rank}(L)) + 1$. On the other hand, by the handshaking lemma, the number of edges is exactly half of $\text{trace}(L)$. So:

G is acyclic if and only if $0.5 * \text{trace}(L) = \text{rank}(L)$. Equivalently, G has a cycle if and only if $0.5 * \text{trace}(L) > \text{rank}(L) + 1$.

edited Apr 23 at 8:48

answered Aug 27 '14 at 21:13



Casteels

141 3

Asotsky's answer is correct in case of directed graphs only. – Pushpendre Apr 23 at 6:38

Casteels' statement that a graph is acyclic iff it has exactly $n-c$ edges only holds in undirected graphs. The statement that G is acyclic if and only if the number of edges is at least $n - (n - \text{rank}(L)) + 1$ is obviously wrong (at least should be at most perhaps) – Pushpendre Apr 23 at 6:52

@Pushpendre The original question was about undirected graphs. However upon retrospect, it is unclear if OP is just looking for triangles or cycles of any length. The definition of cycle she is using seems to imply she wants triangles, but the proposed algorithm makes it appear she is looking for any length cycle. Anyways, even for directed graphs, Asotsky's answer is incorrect since any undirected graph can be made directed by replacing each edge by two directed edges, one going in either direction. So my counterexample still works. But yes, there is a typo in that sentence, thanks. – Casteels Apr 23 at 8:48

Let's use the "counterexample" you used. A path of length 4 would contain a 4-cycle according to Danil's answer. The adjacency matrix in a directed graph of a 4-path to the power 4 would be $\{\{0, 1, 0, 0\}, \{0, 0, 1, 0\}, \{0, 0, 0, 1\}, \{0, 0, 0, 0\}\}^4$. Anyone can confirm that this does not have diagonal entries by pasting in wolfram alpha. You are right that an undirected graph can be converted into directed one by replacing one edge by two edges, but doing so automatically adds cycles to the graphs so this construction is useless. – Pushpendre Apr 24 at 5:23

@Pushpendre My point is that if Danil's answer was correct for directed graphs, then it would be correct for undirected graphs as well, which it is not. The counterexample in my previous comment does not have the adjacency matrix you wrote down; I said to replace each edge with a directed edge in each direction. This gives the same adjacency matrix as the undirected case. Are you sure you are not confusing cycle with closed walk? – Casteels Apr 24 at 9:20

This approach uses DFS, but is very efficient, because we don't repeat nodes in subsequent DFS's.

High-level approach:

Initialize the values of all the nodes to -1 .

Do a DFS from each unexplored node, setting that node's value to that of an auto-incremented value starting from 0 .

For these DFS's, update each node's value with $\text{previous node's value} + i/n^k$ where that node is the i th child of the previous node and k is the depth explored, **skipping already explored nodes** (except for checking for a bigger value).

So, an example for $n = 10$:

```

0.1  0.11  0.111
 j   - k   - p
0 /   \ 0.12
i \ 0.2  1
   m

1  1.1
q - o
...
```

You can also use $i/\text{branching factor}+1$ for each node to reduce the significant digits of the numbers, but that requires additional calculation to determine.

So above we did a DFS from i , which had 2 children j and m . m had no children, j had 2 children, Then we finished with i and started another DFS from the next unexplored node q .

Whenever you encounter a bigger value, you know that a cycle occurred.

Complexity:

You check every node at most once, and at every node you do n checks, so complexity is $O(n^2)$, which is the same as looking at every entry in the matrix once (which you can't do much better than).

Note:

I'll also just note that an [adjacency list](#) will probably be faster than an adjacency matrix unless it's a very dense graph.

edited May 8 '13 at 11:16

answered May 8 '13 at 10:05



Dukeling

32.8k 8 31 69

That is the problem I also found. The explanation, I thought, is the following:

when we talk about cycle, implicitly we mean directed cycles. The adjacency matrix that you have has a different meaning when you consider the directed graph; it is indeed a directed cycle of length 2. So, the solution of A^n is actually for directed graphs. For undirected graphs, I guess a fix would be to just consider the upper triangular version of the matrix (the rest filled with zero) and repeat the procedure. Let me know if this is the right answer.

edited Jan 8 '14 at 14:57

answered Jan 8 '14 at 14:32



Olimpiu POP

2,236 12 35



Swaprava

8 3

If digraph G is represented by its Adjacency matrix M then $M'=(I - M)$ will be singular if there is a cycle in it. I : identity matrix of same order of M

answered Oct 27 '14 at 17:52



Saptarshi Ghosh

1

1 This is false: Let G be the directed graph with adjacency matrix $M = \{\{0,1,0,0\},\{0,0,1,1\},\{1,0,0,1\},\{1,0,0,0\}\}$. This has directed cycles but $I-M$ is nonsingular ($\det(I-M)=-2$) – [Casteels](#) Jan 19 at 9:36

Some more thoughts on the matrix approach... The example cited is the adjacency matrix for a disconnected graph (nodes 1&2 are connected, and nodes 3&4 are connected, but neither pair is connected to the other pair). When you calculate A^2 , the answer (as stated) is the identity matrix. However, since $\text{Trace}(A^2) = 4$, this indicates that there are 2 loops each of length 2 (which is correct). Calculating A^3 is not permitted until these loops are properly identified and removed from the matrix. This is an involved procedure requiring several steps and is detailed nicely by R.L. Norman, "A Matrix Method for Location of Cycles of a Directed Graph," AICHE J,

11-3 (1965) pp. 450-452. Please note: it is unclear from the author whether this approach is guaranteed to find ALL cycles, UNIQUE cycles, and/or ELEMENTARY cycles. My experience suggests that it definitely does not identify ONLY unique cycles.

[edited Oct 27 '14 at 23:47](#)

[answered Oct 26 '14 at 3:20](#)



[Don Elmore](#)

1 1
